Topics:

- Limits of functions
- Limits and algebraic operations
- Continuity
- Continuity and composition

Def Let
- X, Y be metric spaces,
- $E \subset X$ be a subset,
- $f : E \to Y$ be a function,
- $p \in X$ be a limit pt of E,
- and let $q \in Y$.

We write $f(x) \to q$ as $x \to p$ or

$$\lim_{x \to p} f(x) = q.$$

If for every $\varepsilon > 0$, there exists $\delta > 0$ s.t.,

$0 < d_X(x, p) < \delta$ implies $d_Y(f(x), q) < \varepsilon$ for all $x \in E$.

Example Let $f(x) = 5x$.

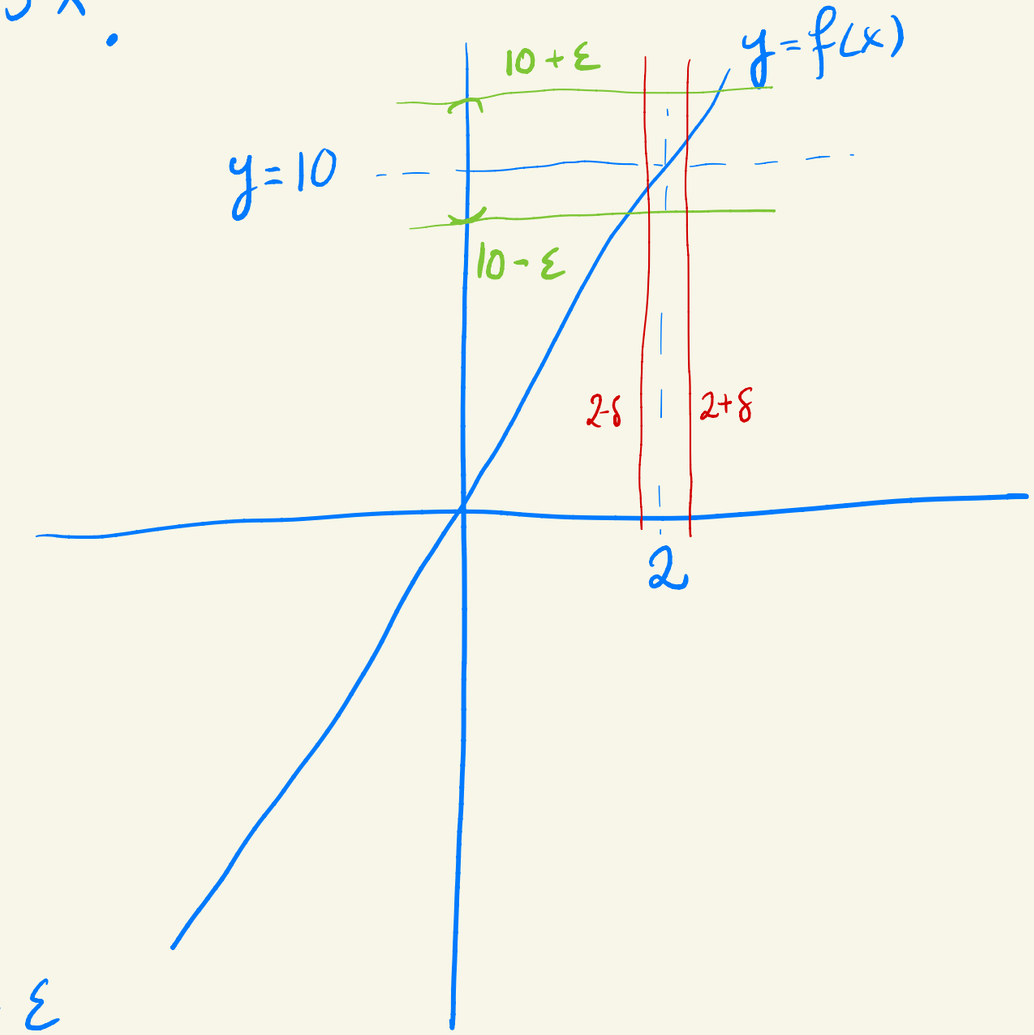
$$\lim_{x \to 2} f(x) = 10$$

Proof: Let $\varepsilon > 0$.

Let $\delta = \frac{\varepsilon}{5}$.

If $|x-2| < \delta$, then

$$|f(x)-10| = |5x-10| = 5|x-2| < 5\delta = \varepsilon$$



Example/Exercise Consider the functions $\mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$

$$f(x,y) = \frac{x^2 - y^2}{x^2 + y^2} \qquad g(x,y) = \frac{x^2 - y^2}{\sqrt{x^2 + y^2}}$$

$\lim_{(x,y)\to(0,0)} f(x,y)$ doesn't exist

$\lim_{(x,y)\to(0,0)} g(x,y) = 0$

Thm (4.2) Suppose $E \subset X$, $p \in E'$, $f: X \to Y$.

TFAE:

① $\lim_{x \to p} f(x) = q$

② $\lim_{n \to \infty} f(p_n) = q$ for every seq (p_n) in $E \setminus \{p\}$ with $\lim_{n \to \infty} p_n = p$.



Proof: ① ⇒ ② Let $\{p_n\}$ be a seq in $E \setminus \{p\}$ with $p_n \to p$.

Let $\varepsilon > 0$. Let $\delta > 0$ be s.t. $0 < d(x,p) < \delta \Rightarrow d(f(x), q) < \varepsilon$

$\forall x \in E \setminus \{p\}$. Let N be s.t. $n \geq N \Rightarrow d(p_n, p) < \delta$.

If $n \geq N$ then $d(f(p_n), q) < \varepsilon$.

② → ① Suppose ① false: $\exists\ \varepsilon>0$ s.t. $\forall\ \delta>0$ there exists $x \in E \setminus \{p\}$ s.t. $d(x,p)<\delta$ and $d(f(x),q) \geq \varepsilon$. Fix such an $\varepsilon>0$, and let $p_n \in E$ be a point s.t. $0 < d(p_n,p) < \frac{1}{n}$ and $d(f(p_n),q) \geq \varepsilon$. Then $f(p_n)$ does not go to q.

Corrolary $\lim_{x\to p} f(x) = q$ and $\lim_{x\to p} f(x) = q' \Rightarrow q=q'$

Suppose $p_n \to p$. Then $f(p_n) \to q$ and $f(p_n) \to q'$, so $q=q'$.

Corrolary Say $E \subset X$, $f,g : E \to \mathbb{R}$, and $f(x) \to A$, $g(x) \to B$ as $x \to p$. Then as $x \to p$,

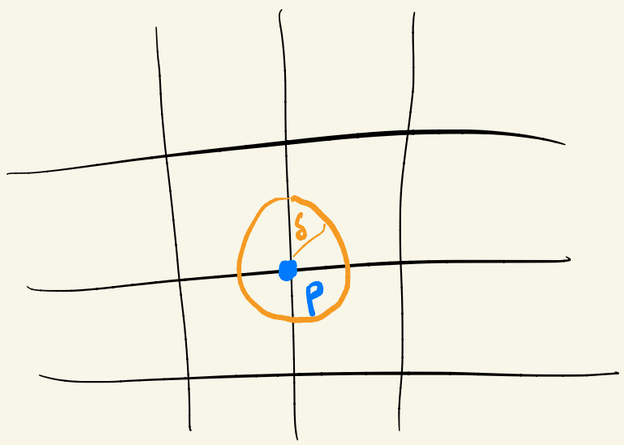
① $f+g \to A+B$

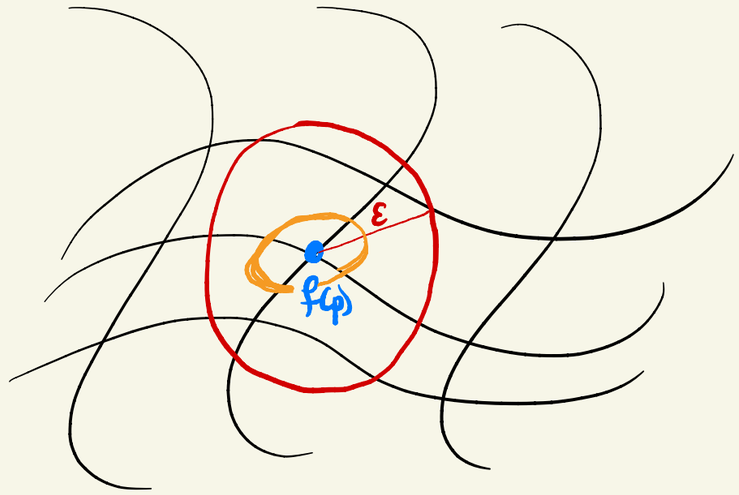
② $fg \to AB$

③ $\frac{f}{g} \to \frac{A}{B}$ if $B \neq 0$.

For any $p_n \to p$, we have $f(p_n) \to A$ and $g(p_n) \to B$.

Def A map $f: X \to Y$ of metric spaces is <u>continuous at p</u> if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $d(p,q) < \delta \implies d(f(p), f(q)) < \varepsilon$.





f is <u>continuous</u> if it is continuous at all $p \in X$.

Equiv: $f(N_\delta(p)) \subset N_\varepsilon(f(p))$

Examples

- Identity $X \to X$ is continuous (take $\delta = \varepsilon$)
- $f(x) = 2x + 3$ is continuous (take $\delta = \frac{\varepsilon}{2}$)
- Any $f : \mathbb{N} \to \mathbb{R}$ is continuous (take $\delta = \frac{1}{2}$)
- Fix $q \in X$. $f(p) = d(p, q)$ is continuous.

Thm (4.6) ① Suppose $p \in X$ is a limit pt of X.

$f: X \to Y$ is continuous at p iff $\lim_{x \to p} f(x) = f(p)$.

② Suppose $p \in X$ is not a limit point of X.

Then any $f: X \to Y$ is continuous at p.

Proof: Exercise

Cor If $f, g: X \to \mathbb{R}$ are continuous, then so are

$f+g$, fg, and $\frac{f}{g}$ if g is non-vanishing.

Thm Suppose $f: X \to Y$ and $g: Y \to Z$.

If f is cont. at p and g is cont. at $f(p)$, then

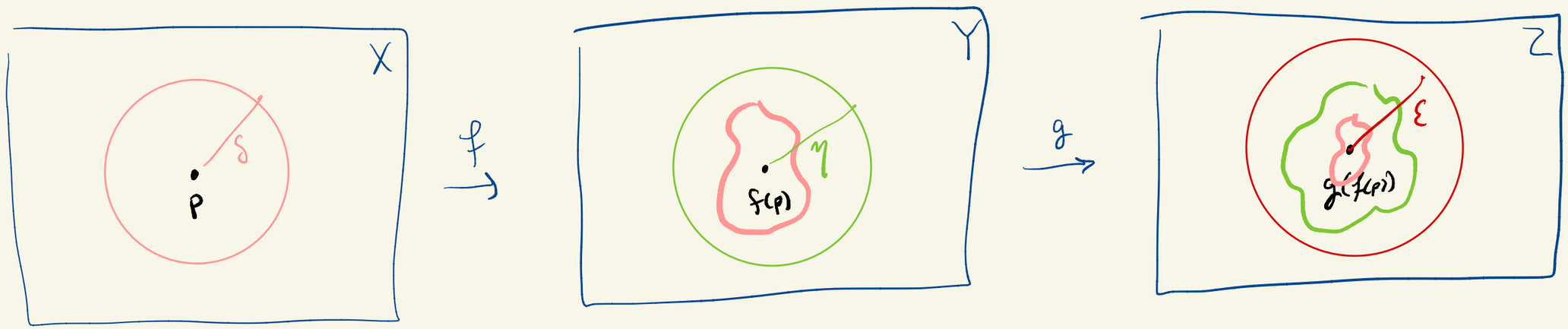
$g \circ f$ is cont. at p.

proof: Let $\varepsilon > 0$.

$\exists\, \eta > 0$ s.t. $g\left(N_\eta(f(p))\right) \subset N_\varepsilon(g(f(p)))$.

$\exists\, \delta > 0$ s.t. $f(N_\delta(p)) \subset N_\eta(f(p))$.

We have $g(f(N_\delta(p))) \subset N_\varepsilon(g(f(p)))$



Example: At what points are the following functions $\mathbb{R}\to\mathbb{R}$ continuous:

① $f(x)$ = a polynomial in x

② $$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ 1 & x \notin \mathbb{Q} \end{cases}$$

③ $$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 1 & x \notin \mathbb{Q} \end{cases}$$

④ $$f(x) = \begin{cases} 1 & \text{if} \quad x \in \mathbb{N} \\ \frac{1}{2^k} & \text{if} \quad x = \frac{n}{2^k} \text{ for } k \in \mathbb{N}, \text{ and odd } n \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

① use

- $f(x) = x$ is continuous
- $f(x) = c$ is continuous for any $c \in \mathbb{R}$. (any δ works for any ε)
- f, g continuous $\Rightarrow fg$ and $f+g$ continuous

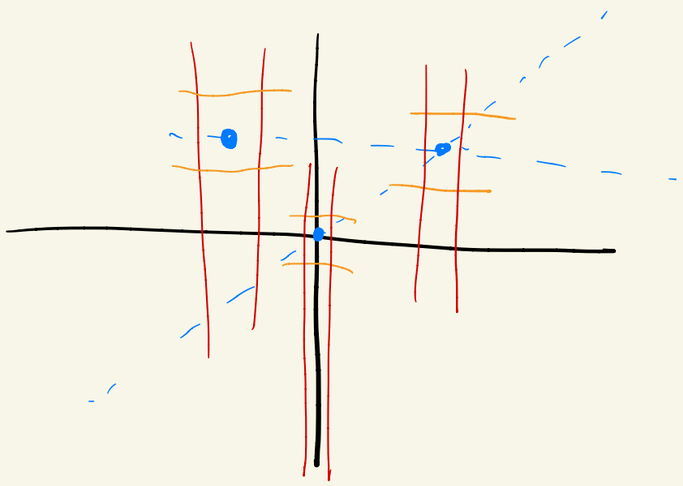
② Nowhere continuous. Suppose $\varepsilon = \frac{1}{2}$.

Let $p \in \mathbb{R}$. Let $\delta > 0$.

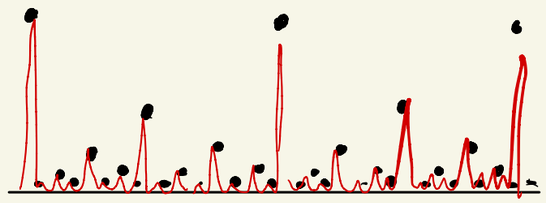
If $p \in \mathbb{Q}$ then $\exists\ q \in (p-\delta, p+\delta) \setminus \mathbb{Q}$ and $|f(p) - f(q)| = 1 > \varepsilon$

If $p \in \mathbb{R} \setminus \mathbb{Q}$ then $\exists\ q \in (p-\delta, p+\delta) \cap \mathbb{Q}$ and $|f(p) - f(q)| = 1 > \varepsilon$.

③



④



Is the identity continuous $(\mathbb{R}, d_{disc}) \longrightarrow (\mathbb{R}, d_{std})$?

yes. For any $\varepsilon > 0$, choose $\delta = \frac{1}{2}$ so that $N_\varepsilon^{disc}(x) = \{x\}$ for all x.

What about as a function $(\mathbb{R}, d_{std}) \longrightarrow (\mathbb{R}, d_{disc})$?

no. For $\varepsilon = \frac{1}{2}$, $N_\varepsilon^{disc}(x) = \{x\}$ so no $N_\delta^{std}(x) = (x-\delta, x+\delta)$ fits inside.